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Separated Ionic-Electronic Conduction in Hydrophobic Conjugated Polymer/Hydrophilic Photocrosslinker Blends for Organic Electrochemical Transistors

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ABSTRACT

Ionic-electronic coupling serves as the core process enabling the operation of organic mixed ionic-electronic (semi)conductors (OMIECs) based devices, for instance, organic electrochemical transistors (OECTs). Replacing hydrophobic side chains of conjugated polymers with hydrophilic ethylene glycol/ionic ones is a well-developed approach to enable transistor channels with coupled ionic and electronic transport. Here, in contrast, we introduce a hydrophilic glycol chain-modified photocrosslinker (DtFGDA) for the direct photolithography process and blend it with various representative hydrophobic conjugated polymers. The precise patterning of blended films by direct photolithography is achieved while tremendous enhancements of OECTs performance are attained, with maximum six orders of magnitude higher transconductance, significantly decreased hysteresis, and lower threshold voltage. Through spectroelectrochemical characterization, surprisingly, no obvious variations in polaron absorption peaks are observed in all conjugated polymer/crosslinker blends. An ionic-electronic separated conduction mechanism, which is never reported in OECTs before, is further proposed based on the characterization of the transmission electron microscope, wherein ions primarily migrate within the crosslinker while holes transport within the semiconducting polymer. This work proposes an efficient strategy, which involves incorporating hydrophilic chains into the photocrosslinker necessary for direct photolithography and blending it with hydrophobic semiconducting polymers, achieving synergistic ionic-electronic transport in the blended film.

1 | Introduction

According to existing high-performance hydrophobic polymer materials for organic semiconductor devices [1–5], a side-chain modification that replaces alkyl side chains with ethylene glycol/ionic ones is one of the pathways to obtain novel organic mixed ionic-electronic (semi)conductors (OMIECs) [6–10]. However, due to the variation of side chain type [11, 12], the introduction of ethylene glycol/ionic side chain, both the solubility and energy levels of conjugated polymers would vary [13–15]. To obtain a polymer with improved performance, additional synthesis steps are required to accomplish further

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modification of functional groups or alteration of bridging groups [13, 16, 17]. This would increase the research costs for high-performance material synthesis. As the major research tool and applicable electronic device of OMIECs, organic electrochemical transistor (OECT) is switched by the gate voltage, where ions from the electrolyte can inject/eject to/from the channel, altering the redox state of it and modulating the channel current [18–21].

Organic semiconductors have achieved remarkable advancements over the past decades [22, 23]. Unfortunately, most representative semiconducting polymers with efficient charge transport are usually hydrophobic, which prohibits efficient electrochemical doping, making them unsuitable to act as OMIECs. To enable efficient mixed ionic-electronic conduction in traditional hydrophobic organic semiconductors, several methods for constructing ionic pathways have been reported, including forming porous semiconductor film in the channel layer [24], adding ionic exchange gel between the channel and electrolyte [25, 26], and replacing aqueous electrolyte with a hydrophobic ionic liquid ($EMIM^+$ & $TFSI^-$) [27]. Preparation of porous films with hydrophobic polymers poly(3,6-di(-2-thien-5-yl)-2,5-di(2-octyldodecyl)-pyrrolo[3,4-c]pyrrole-1,4-dione) thieno[3,2-b]thiophene (DPPDTT) and poly(3-hexylthiophene-2,5-diyl) (P3HT) using the breath figure method can achieve fast ion injection/ejection, making their performance comparable to that of hydrophilic polymer. By incorporating a layer of ion exchange gel between the active layer and the aqueous electrolyte, the application of a gate bias enables the insertion of ions, such as chloride ions, from the aqueous electrolyte into the ion exchange gel. Subsequently, the hydrophobic $TFSI^-$ ions are inserted into the bottom channel layer to compensate for the holes within the polymer backbone, resulting in excellent performance of OECT. However, both porous films and ionic exchange gel strategies have the issue of making the device process more complex. Hydrophobic P3HT has been directly employed to function as OMIECs, utilizing a hydrogel that contains hydrophobic ions. Nevertheless, high turn-on voltage (V_{on}) (> 1 V) and limits on electrolyte types greatly hinder the application of P3HT-based OECTs. Therefore, the search for a more straightforward and effective method for constructing ionic pathways within typical hydrophobic polymers will further advance the field of OMIECs. Currently, direct photolithography using photocrosslinker is one of the most efficient approaches for precise patterning in organic electronics. Considering this, we propose modifying the structure of photocrosslinkers by introducing hydrophilic chains to achieve patterning while constructing ionic transport channels in hydrophobic semiconductor films. This approach lowers requirements for OMIECs materials.

Here, an ethylene glycol-based double-end trifluoromethyl aryl diazirine (DtFGDA) photocrosslinker is blended with four commonly used p-type hydrophobic polymers, P3HT [28–30], poly(3,3‴-didodecyl quaterthiophene) (PQT) [31], poly[thieno [3,2-b]thiophene–2,5-diyl(4,4'-ditetradecyl[2,2'-bithiophene]-5,5'-diyl)] (PBTTT) [32], and DPPDTT [33] utilized as the channel of vertical OECTs (vOECTs) (Figure 1A) [34, 35]. Such blends enable efficient ion-electron synergistic conduction where peak transconductances ($g_{m,peak}$) for vOECTs based on P3HT, PQT, PBTTT, and DPPDTT reach 131.0, 160.0, 230.9, and 10.4 mS, respectively, which are comparable to those reported based on state-of-the-art OMIECs [14, 36, 37]. Furthermore, significant reductions in switching time and hysteresis are also observed. Turn-on transient time (τ_{on}) of P3HT, PQT, PBTTT, and DPPDTT-based vOECTs are reduced to 79.1, 215.8, 94.8, and 363.3 ms, respectively. Remarkably, in the characterization of spectroelectrochemistry (SEC) [38, 39], no obvious changes in the π-π* absorption and polaron absorption of hydrophobic polymers were observed with the addition of photocrosslinker. It indicates that under bias voltage, the hydrophobic conjugated polymers are almost not electrochemically doped. However, the electrochemical quartz crystal microbalance (EQCM) [40, 41] characterization reveals the blended film is doped/de-doped by ions. Through transmission electron microscope (TEM), significant phase separation is identified in the blended film, which consists of domains of hydrophilic crosslinker and hydrophobic polymer. Therefore, we assume that the ions are mainly doped into the hydrophilic

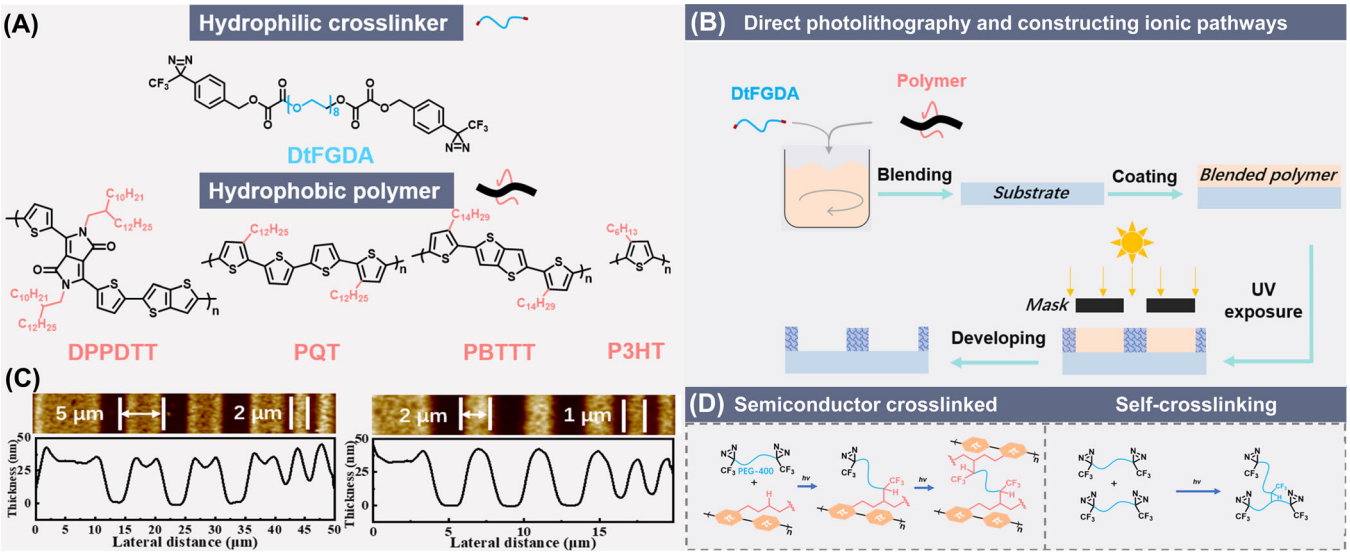


FIGURE 1 | (A) Chemical structures of DtFGDA and polymer. (B) The process of using DtFGDA for direct photolithography. (C) AFM images of the patterned thin films of PBTTT: DtFGDA (left) and DPPDTT: DtFGDA (right). (D) Two types of crosslinking occur in thin films.

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crosslinker domains and lead to the formation of field-effect doping within the crosslinker-polymer interface. This doping mechanism differs from the well-established doping form of OECTs [7]. This is an ionic-electronic separated conduction mechanism, where ions are transported through hydrophilic crosslinkers and electrons are transported through hydrophobic polymers. Overall, these results demonstrate that our strategy successfully constructs ionic transport channels in hydrophobic polymers and provides guidelines for the future design of high-performance OMIECs.

2 | Experimental Section/Methods

2.1 | Materials

All reagents and chemicals were purchased from commercial sources and were used without further purification. 4-[3-(Trifluoromethyl)-3H-diazirin-3-yl]benzyl alcohol (96%) was purchased from Bidepharm. Oxalyl chloride (98%) was purchased from chronchem. Poly(ethylene glycol) (average M_n 400) was purchased from Macklin. 5,5′-dibroMo-4,4′-didodecyl-2,2′-bithiophene was purchased from aladdin. 5,5′-bis(triMethylstannyl)-2,2′-bithiophene,2,5-bis(triMethylstannyl)-thieno[3,2-b]thiophene and 4,4′-Dibutoxy-2,2′-bis(trimethylstannyl)-5,5′-bithiazole was purchased from SunaTech Inc. $Pd_2(dba)_3$ was purchased from Sigma-Adrich. P(*o*-tol)$_3$ was purchased from Adamas-beat. 5,5′-dibroMo-4,4′-ditetradecyl-2,2′-bithiophene was purchased from Zhi-yan. Detailed synthesis routes for the DtFGDA and the polymers (P3HT, PQT, PBTTT, DPPDTT) are provided in the supporting information (Supporting Information S1: Schemes 1–4).

2.2 | Fabrication Process of Vertical OECTs

The vertical OECTs (vOECTs) were fabricated following the procedures in the reported literature [34]. Semiconductor solution preparation: The conjugated polymer was stirred in chlorobenzene at 80°C for 6 h to prepare a solution with a concentration of 20 mg/mL. The crosslinkers in chlorobenzene were stirred at room temperature for 0.5 h to prepare a solution with a concentration of 100 mg/mL. Then, the polymer was blended with the crosslinker at a weight ratio of 1:4 and stirred in chlorobenzene at 60°C for 2–3 min to fabricate the channel layer. Encapsulation layers are fabricated according to the procedures in the reported literature [35]. The vOECT devices were fabricated on a Si/SiO_2 wafer (2.2 × 2.2) which was pre-cleaned ultrasonically with isopropyl alcohol and subsequently treated with plasma. First, 3 nm of Cr and 150 nm of Au were thermally evaporated successively with a shadow mask to form the bottom source electrode on the wafer. Next, the semiconductor (16 mg/mL in chloroform) or blended semiconductor/photocrosslinker solution (20 and 100 mg/mL) was spun-coated on the wafer at 1500 r/min for 1 min as the channel. Then, the channel layer was exposed to UV light through a photomask for 3 min and was developed in chloroform for 3 s to complete the patterning (The pure polymer channel layer has not a patterning process). The top drain electrode (150 nm Au) was then thermally evaporated with a shadow mask while maintaining the wafer at a temperature of approximately 25°C with a cycle water-cooling system. The encapsulation solution was spun coated on the devices at 5000 r/min for 15–20 s. And the encapsulation layer was then exposed to UV light through a photomask for 4 min and developed in chloroform for 3 s and blow-dried (the pure polymer channel layer has not a patterning process.). Finally, a droplet (approximately 1–2 μL, based on the channel area) of 0.01 mol/L PBS was dripped on the channel area, and an Ag/AgCl electrode was inserted into the droplet as the gate electrode.

2.3 | UV-vis Spectrum

UV-vis absorption spectra measurements were conducted on a UNICO UV–4802 UV/vis spectrophotometer. Ultraviolet-visible spectrum of crosslinkers: DtFGDA were dissolved in chloroform and configured into 100 mg/mL solution, which was added into quartz cuvette for wavelength scanning and normalization. The solution was spun coated on the quartz substrate at 3000 r/min for 15–20 s for wavelength scanning and normalization. Ultraviolet-visible spectrum of crosslinked film: the blended polymer solution is prepared in the same way as the vOECT fabrication part. The solution was spun coated on the quartz substrate at 3000 r/min for 15–20 s, and then exposed for 3 min. (Light source: 365 nm, intensity: 600–800 mW/cm^2).

2.4 | Atomic Force Microscope

The blended solution is prepared the same way as the device fabrication process. The solution was spun coated on the Si substrate (1 cm × 1 cm) at 1500 r/min for 50–60 s. The thin film was exposed to UV light through a photomask for 3 min. AFM images were recorded using Bruker INNOVA in tapping mode under ambient conditions.

2.5 | Electrochemical Quartz Crystal Microbalance-Dissipation

EQCM-D measurements were performed using a gold-coated quartz crystal sensor and a quartz crystal microblance (QCM922A, Princeton Applied Research). The P3HT (8 mg/mL in chloroform) or blended semiconductor/photocrosslinker solution (8 mg/mL and 100 mg/mL) was spun-coated on the wafer at 1500 r/min for 1 min, followed by exposure to UV light (365 nm) for 3 min. Immediately after the stabilization of the polymer-coated sensor in the electrolyte solution (0.1 mol/L PBS), the system was connected to a PRIMARIUS MODEL: FS373 for biasing the film. The three-electrode setup comprised an Ag/AgCl reference electrode, a Pt counter electrode, and the polymer-coated sensor as the working electrode. In the measurements, four cycles of the 0 V → 0.8 V → 0 V bias sequence were applied. Sauerbrey equation was used to convert $\Delta f/n$ to the change in areal mass (Δm):

$$\Delta m = -17.7\frac{\Delta f}{n}$$

2.6 | Spectroelectrochemistry

The blended polymer solutions were spun coated onto ITO glass and then exposed to UV light at 365 nm wavelength for 3 min.

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The samples were then immersed in a cuvette filled with electrolyte (0.01 mol/L, PBS) and connected to an Ag/AgCl electrode. In-situ monitoring of the UV-vis-NIR spectra of the pure/blended polymer films was conducted at various voltages (vs Ag/AgCl) to investigate their neutral and charged states during electrochemical doping using a KEITHLEY 2401 Source Meter. A UV-3600i Shimadzu spectrophotometer analyzed the light passing through the cuvette containing the electrolyte and the polymer/ITO glass sample. A background spectrum was recorded before the experiment for baseline correction. After applying a potential for 5 s, the UV-visible spectrum was obtained.

2.7 | Transmission Electron Microscope and Energy Dispersive Spectroscopy

PSS solution is first spun on the quartz sheet (2 cm × 2 cm) at 3000 r/min for 15–20 s, then the blended solution was spun coated on it at 3000 r/min for 1 min. After exposure for 3 min, the film was soaked in deionized water and carefully scooped up with a copper mesh. TEM and EDS measurements were performed using Talos F200S and SUPER X, respectively.

3 | Results and Discussion

3.1 | Materials Preparation and Direct Photolithography

The DtFGDA is synthesized through two straightforward substitution reactions with high yields (99%). Benzyl alcohol containing diazirine groups is added dropwise to an excess of oxalyl chloride to form an intermediate product. Excess oxalyl chloride and solvent are removed by a vacuum rotary evaporator. Detailed synthesis routes for the crosslinker and the polymers are provided in the supporting information (Supporting Information S1: Schemes 1–4). After blending DtFGDA with the polymer, the patterning of the channel is accomplished in only three steps: spin coating, exposure, and development (Figure 1B) [42]. The direct photolithography approach is more efficient than conventional photolithography techniques which typically require a minimum of six steps in organic transistors [43]. Since DtFGDA crosslinks with the polymer, it will remain in the blended film after development and form a hydrophilic ionic transport pathway. When exposed to ultraviolet (UV) light, diazirine releases nitrogen gas, immediately followed by the formation of carbenes, which then insert into sp^3 CH bonds to complete the crosslinking process [44, 45]. This high reactivity of the crosslinking reaction enables precise patterning with a resolution as fine as 2 to 5 μm using direct photolithography (Figure 1C). Importantly, the ability to construct ionic pathways while achieving precise patterning broadens the applicability of this approach.

3.2 | Characterization of DtFGDA and Crosslinked Hydrophobic Polymers

Due to the crosslinker itself containing sp^3 CH bonds, two primary types of crosslinking reaction occur in the blended film under UV light exposure at 365 nm (Figure 1D): crosslinking between the crosslinkers and the polymers, as well as self-crosslinking of the crosslinkers. The UV-vis absorption spectra of DtFGDA are shown in Figure 2A and Supporting Information S1: Figure S1, indicating maximum absorption wavelengths of 352 nm for the solution and 355 nm for the film. After exposure, the maximum absorption peak of the DtFGDA at 355 nm disappears. This indicates that the peak corresponds to the n-π transition of the diazirine group. Following 3 min of exposure, the diazirine in the film undergoes complete reaction, releasing nitrogen and completing the self-crosslinking process. The UV-vis absorption spectra of the crosslinked blended film reveal that, following crosslinking reaction, the 0-0 peaks (ground state to first excited state) of all polymers are enhanced (P3HT at 607 nm, PQT at 585 nm, PBTTT at 584 nm, and DPPDTT at 831 nm, respectively) (Figure 2B–E and Supporting Information S1: Figure S2). Additionally, the maximum absorption wavelengths of P3HT and DPPDTT show redshifts of 42 and 19 nm, respectively. This redshift and the enhancement of the 0-0 peaks are probably attributable to the selective swelling of excessive DtFGDA. It leads to increased repulsive forces between excessive DtFGDA and conjugated polymers, which may in turn induce stronger intrachain exciton coupling [46]. In contrast, the overlap of ethylene glycol chains results in no obvious shoulder peaks or redshifts being observed in the films (Figure 2A). It suggests that DtFGDA weakens the aggregation of conjugated polymers, potentially facilitating the establishment of ionic transport pathways.

Atomic force microscopy (AFM) was used to investigate the surface morphology, roughness, and phase separation of crosslinked films (Figure 2F–I). For the four polymers studied, roughness increased significantly after blending and crosslinking. Initially, the root-mean-square roughness (σ_{rms}) of the neat films for P3HT, PQT, PBTTT, and DPPDTT were 8.18, 1.70, 2.63, and 0.64 nm, respectively, indicating relatively smooth surfaces. After blending and crosslinking, the σ_{rms} of blended films changed markedly, reaching 27.0, 6.67, 23.4, and 34.0 nm, respectively. This significant change in morphology is attributed to the phase separation between the crosslinker and the polymer. The crosslinked films of P3HT and PBTTT exhibit an island-like phase separation morphology, whereas the films of PQT and DPPDTT show a fibrous phase separation morphology. These observations are consistent with the results from subsequent TEM characterization.

To investigate the impact of DtFGDA on the crystal packing of polymers, grazing incidence X-ray diffraction (GIXRD) measurements are used to determine the solid-state packing structure of DtFGDA blended with various polymers, as illustrated in Figure 3. The P3HT neat film exhibits an out-of-plane layered structure with diffraction peaks at $q_z = 0.372\ Å^{-1}$ (100), $0.747\ Å^{-1}$ (200), $1.12\ Å^{-1}$ (300), and $q_{xy} = 1.63\ Å^{-1}$ (010), corresponding to d-spacings of 17.0 Å, 8.41 Å, 5.61 Å, and 3.85 Å, respectively (Figure 3A–F). No distinct lattice diffraction peaks were observed in the in-plane direction, indicating an "edge-on" orientation of the P3HT. As the proportion of DtFGDA increased and after crosslinking, the diffraction intensity and preferred orientation of the P3HT films progressively weakened. With increasing DtFGDA ratio from 1:1 to 1:4, the diffraction peaks at (200), (300), and (010) ($q_z = 0.747\ Å^{-1}$, $1.12\ Å^{-1}$, $q_{xy} = 1.63\ Å^{-1}$) gradually disappeared. Moreover, the (100) diffraction

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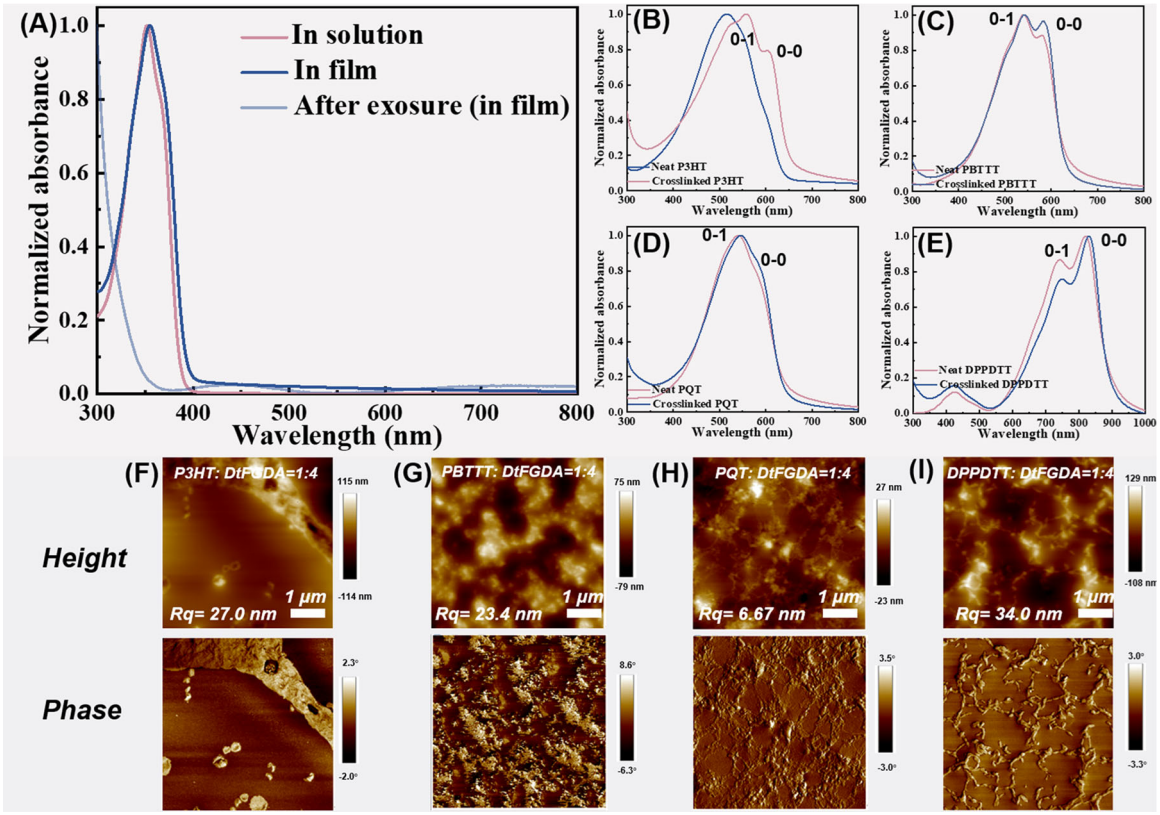


FIGURE 2 | (A) Absorption spectra of DtFGDA in chloroform, film state and film after exposure (365 nm, exposure 3 min). Absorption spectra of the P3HT: DtFGDA (B), PBTTT: DtFGDA (C), PQT: DtFGDA (D), and DPPDTT: DtFGDA (E) films after exposure to UV light (365 nm) for 3 min. AFM height and phase images of the P3HT: DtFGDA (F), PBTTT: DtFGDA (G), PQT: DtFGDA (H), and DPPDTT: DtFGDA (I) films. The weight ratios of polymer: DtFGDA in the blended films are 1:4.

peak ($q_z = 0.372$ Å $^{-1}$) exhibited a broadening of the full width at half maximum (FWHM), increasing from 0.062 Å (neat film) to 0.069 Å (1:1), 0.080 Å (1:2), and 0.088 Å (1:4). The coherent length decreased from 90.797 Å (neat film) to 82.240 Å (1:1), 70.916 Å (1:2), and 64.085 Å (1:4) (Figure 3G). These changes in crystal structure suggest that an increased DtFGDA content may be inserted between P3HT crystals, disrupting the layered packing and leading to a mixed orientation. The crosslinker enhances ionic transport in the film, addressing the issue of ion injection in hydrophobic materials. The same disruption of polymer stacking by the crosslinker is observed in other hydrophobic polymers. For PBTTT, the out-of-plane diffraction peak (400) ($q_z = 1.12$ Å^{-1}) disappears upon the addition of DtFGDA, while in-plane peaks (100) and (200) ($q_{xy} = 0.282$ Å^{-1} and 0.572 Å^{-1}) emerge (Figure 3H, J). This change indicates a transition from an out-of-plane layered structure to a mixed orientation. For PQT, after crosslinking, the FWHM of the (100) peak ($q_z = 0.258$ Å^{-1}) increases from 0.076 to 0.084 Å, and the coherent length decreases from 74.632 to 67.192 Å, while the π-π stacking peak (010) ($q_{xy} = 1.51$ Å^{-1}) vanishes (Figure 3H, I). For DPPDTT, the out-of-plane peak (300) ($q_z = 0.332$ Å^{-1}) disappears, and the FWHM of the (100) peak ($q_z = 0.222$ Å^{-1}) broadens from 0.050 to 0.051 Å, accompanied by a reduction in coherent length (Figure 3K and Supporting Information S1: Table 1). These observations suggest that using ethylene glycol photocrosslinkers to disrupt polymer stacking is an effective strategy for increasing the ionic transport pathways within the film.

3.3 | Organic Electrochemical Transistors Properties of Hydrophobic Polymers

Based on the above characterizations, the crosslinking of DtFGDA with various conjugated polymers enables precise patterning and unique microstructure manipulation. It also reduces the crystalline packing of the material and enhances the hydrophilicity of the films. Consequently, channels based on hydrophobic polymers and DtFGDA are utilized for the fabrication of OECTs (Figure 4A). As the ratio of hydrophobic polymer: DtFGDA changes from 1:1 to 1:4, the hysteresis of the devices reduces (Supporting Information S1: Figure S3). Therefore, a 1:4 ratio of the polymer:DtFGDA is selected for the studied system. Notably, the crosslinker is less expensive than the polymer, offering potential economic benefits when used in high proportions in mixtures. For each vOECT, at least six devices were measured and analyzed, with the corresponding transistor parameters summarized in Table 1, Figure 4, and Supporting Information S1: Figure S4. The $g_{m,peak}$ was extracted from the transfer curves. Hysteresis was quantified by calculating the voltage difference based on the intermediate current between the on and off states.

For the four typical hydrophobic polymers, crosslinking and patterning of the channel layer with DtFGDA resulted in significantly different transfer characteristics. For P3HT (Figure 4B, F), the $g_{m,peak}$ of the pure P3HT-based vOECT was 69.5 ± 21.0 mS, with a switching ratio (I_{on}/I_{off}) of 10^4.

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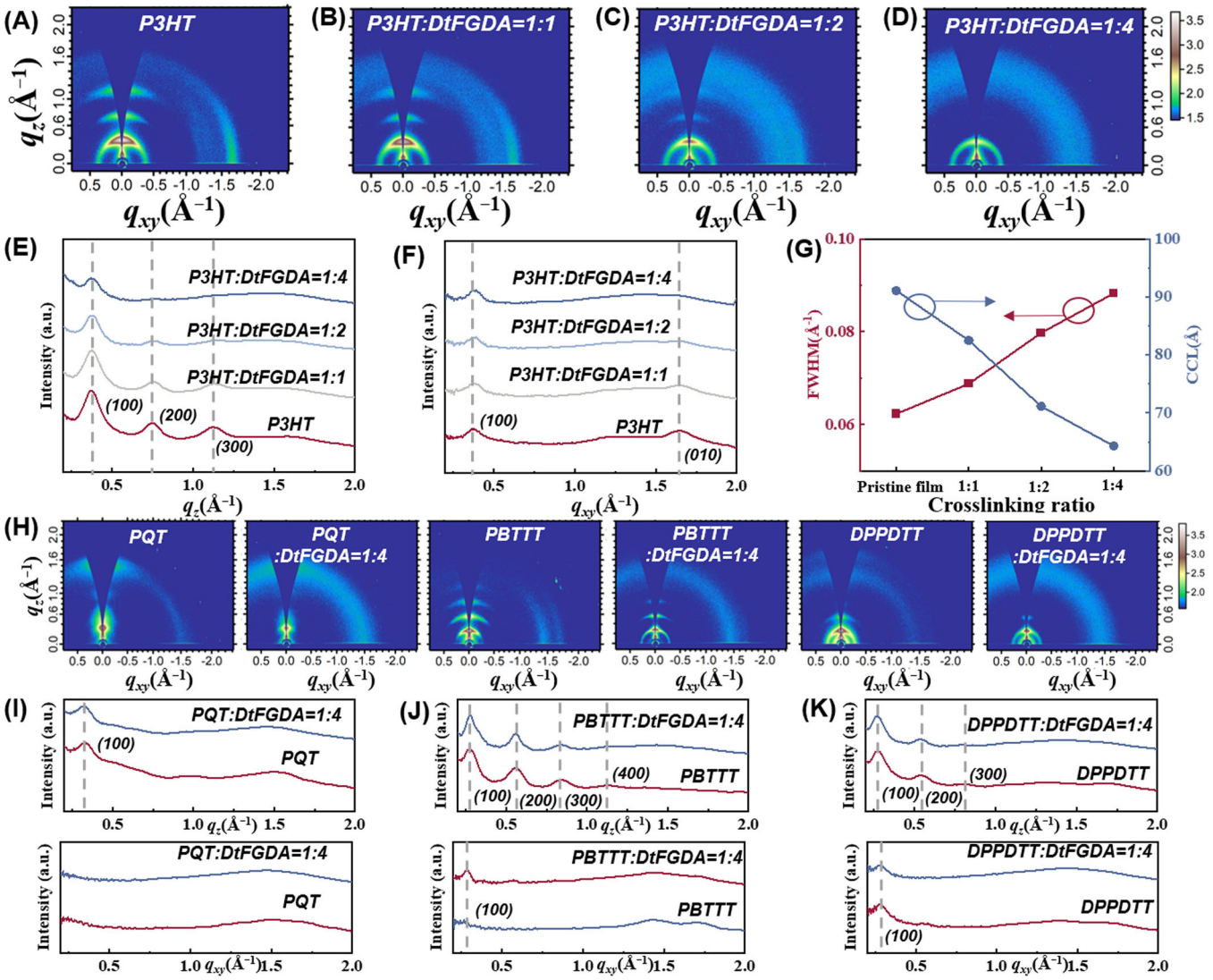


FIGURE 3 | (A–D) 2D GIXRD patterns and (E, F) the linecuts for P3HT: DtFGDA blended films with different blending ratio (G) Full width at half maximum (FWHM) and coherence length (CCL) for DtFGDA: P3HT blended films with different blending ratio. (H) 2D GIXRD patterns for polymer: DtFGDA blended films. Note, polymer: DtFGDA = 1: 4 in mass. (I–K) The linecuts for polymer: DtFGDA blended film.

The device exhibited substantial hysteresis (0.42 ± 0.05 V) and V_{on} of -0.40 ± 0.02 V, indicating a pronounced interface electrostatic operation mode. The hydrophobicity of P3HT hinders ion injection into the film. This static doping creates a 2-dimensional conductive pathway at the interface. In contrast, vOECTs with P3HT:DtFGDA demonstrated significantly improved performance. The $g_{m,peak}$ increased to 131.0 ± 46.9 mS, the I_{on}/I_{off} rose to 10^6, and both hysteresis and V_{on} markedly decreased to 0.04 ± 0.01 and -0.23 ± 0.03 V, respectively. These enhancements are attributed to the crosslinking with DtFGDA, which forms a hydrophilic crosslinked network within the film and disrupts the crystalline packing of P3HT. This modification facilitates easier ionic transport in the film, requires lower voltages for rapid ion injection (evidenced by the reduced hysteresis and operating voltage), and enhances ion-electron coupling. Consequently, conductive pathways form in the film, leading to increased $g_{m,peak,}$ and I_{on}/I_{off}. To further evaluate the channel layer characteristics, turn-on/off transient times were analyzed. Time constants (τ) were estimated through Gaussian fitting. As shown in (Supporting Information S1: Figure S5) and Table 1, the τ_{on} and τ_{off} for P3HT transistor devices were 266.7 ± 15.5 and 250.6 ± 9.1 ms, respectively, which are significantly longer than the transient times of the crosslinked films (τ_{on} of 79.1 ± 4.68 ms and τ_{off} of 5.42 ± 0.99 ms). This demonstrates that ionic transport is slow in hydrophobic P3HT, whereas the crosslinked film with DtFGDA exhibits faster ionic transport. Similar improvements in transfer characteristics were observed in the other three materials: PQT, PBTTT, and DPPDTT (Figure 4C–E). All the materials exhibit more pronounced interface electrostatic operation modes, with $g_{m,peak}$ of 2.2 ± 1.1, 5.9 ± 1.6, and $(1.2 \pm 0.5) \times 10^{-5}$ mS, respectively, accompanied by significant hysteresis and response times. Upon blending and crosslinking with DtFGDA, the $g_{m,peak}$ improved to 160 ± 10.4, 230.9 ± 10.3, and 10.7 ± 6.7 mS, for vOECTs based on PQT, PBTTT, and DPPDTT, respectively (Figure 4G–I). In particular, after crosslinking with DtFGDA, the $g_{m,peak}$ of DPPDTT increased by as much as six orders of magnitude. The I_{on}/I_{off} increased by over 1000 times, and τ_{on} were reduced from 1124.3 ± 51.9, 531.5 ± 27.6, and 7206.4 ± 187.4 ms to 215.8 ± 8.7, 94.8 ± 2.1, and 363.3 ± 17.7 ms,

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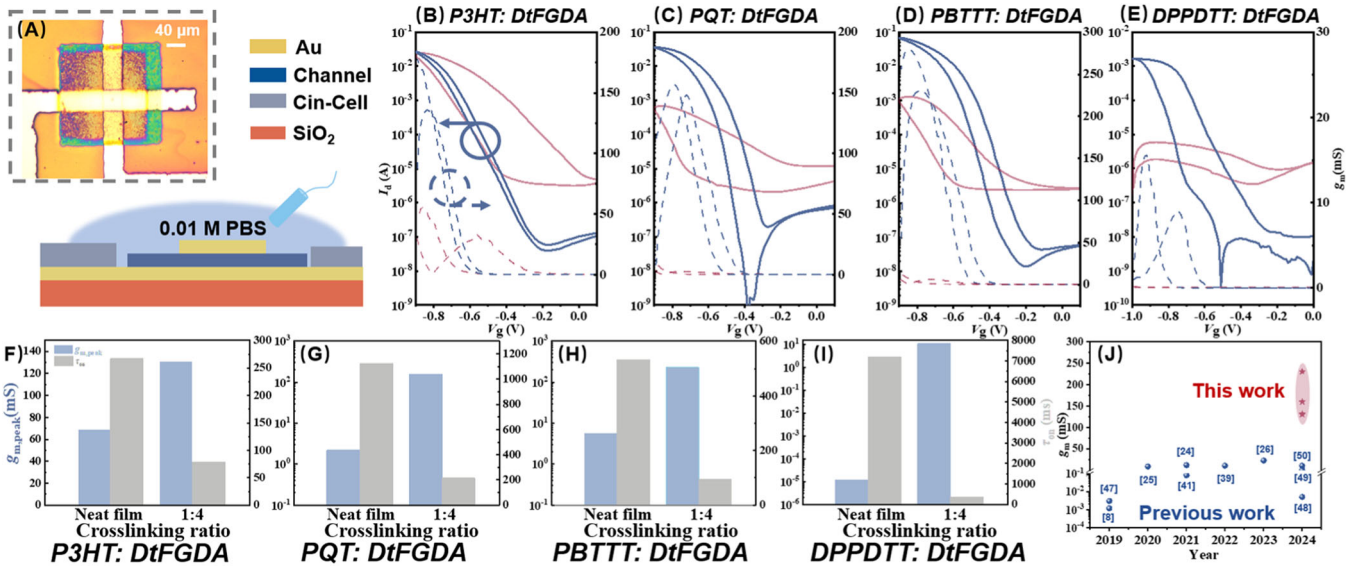


FIGURE 4 | vOECT structure and transistor performance characteristics. (A) Optical image side-view diagrams of vOECT. Transfer (solid line) and transconductance (dotted line) characteristics of P3HT (red) and P3HT: DtFGDA (blue) (B), PQT(red) and PQT: DtFGDA (blue) (C), PBTTT (red) and PBTTT: DtFGDA (blue) (D), DPPDTT (red) and DPPDTT: DtFGDA (blue) (E). Dependencies of g_m (blue) and τ_{on} (gray) on different polymers in vOECTs (F–I). Note, polymer: DtFGDA = 1: 4 in mass. (J) Comparison of $g_{m,peak}$ with those of references [8, 24–26, 39, 41, 47–50].

respectively. The V_{on} and hysteresis were significantly reduced. These results indicate that compared to neat films, vOECTs based on crosslinked films exhibit markedly faster high-current switching characteristics. This improvement is attributed to the formation of a hydrophilic crosslinked network by DtFGDA, which facilitates the injection and transport of phosphate and enhances ion-electron coupling. This represents an effective and versatile strategy for constructing ionic pathways in p-type conjugated polymers. It is the highest performance achieved by hydrophobic polymers (PBTTT with $g_{m,peak}$ of 230.9 ± 10.3 mS) in OECTs to date.

3.4 | Research on Ionic-Electronic Conduction Mechanism

To explore the absorption changes in neat films and blended films of the hydrophobic polymer under applied bias (electrochemical doping) in aqueous electrolytes (0.01 mol/L PBS solution), SEC experiments were conducted using a two-electrode setup. The voltage range for the tests was determined based on the transfer curve measurements. Figure 5 and Supporting Information S1: Figures S6 and S7 present the UV-vis-NIR normalized absorbance spectra for each polymer and blended film under varying applied potentials. Despite the significant performance differences observed in OECT tests, their electrochemical doping and electrochromic properties are comparable. Specifically, for P3HT (Figure 5A, Supporting Information S1: Figure S6), applying −0.9 V on the Ag/AgCl electrode results in limited penetration of hydrophilic anions, such as phosphate and chloride ions, into the film, leading to ineffective doping. This is reflected in the SEC by only a 4.7% change in the π-π* absorption band ($\lambda = 564$ nm) with decreasing voltage, while the polaron absorption peak ($\lambda = 761$ nm) remains nearly constant. Consequently, this results in significant hysteresis and longer response times in OECT performance. After completing crosslinking, the blended films were tested under identical conditions. Remarkably, the π-π* absorption band of the P3HT and DtFGDA blended films exhibited minimal changes ($\lambda = 560$ nm), with only a 8.9% variation after applying −0.9 V to the Ag/AgCl electrode (Figure 5B). The change of the polaron absorption peak was only slightly greater than that observed in the neat film. In contrast, OMIECs such as p(g2T-T) show variations of 50% or more in the π-π* absorption band and polaron absorption peak under applied voltage [51] (Figure 5C, D). Similar to P3HT, the π-π* absorption of PQT, PBTTT, and DPPDTT blended with the DtFGDA changed only by 6.9%, 2.8%, and 1.0%, respectively (Figure 5E, Supporting Information S1: Figure S7). In contrast to OMIECs with ethylene glycol side chains, crosslinked hydrophobic conjugated polymers show little bulk doping. However, its performance is significantly better than typical electrostatic doping. We surmise that this is a unique ion-electron transport mechanism which is different from the mechanism reported so far.

To verify whether the blended films experience the ionic doping/de-doping and hydration processes, EQCM measurements were performed (Figure 5F). As the voltage increased, there was an initial decrease in mass, which could be attributed to the cations and water leaving the film [52]. As the voltage reduced, the mass of the film increased, which is attributed to the rehydration of the film [8]. Notably, the P3HT:DtFGDA blended film is able to return to its initial mass after cycling, while the neat film exhibited significantly irreversible changes in mass. This observation reveals that the injection and ejection of water and ions are hindered in the neat films, whereas they are promoted in the photocrosslinked films.

Morphological analyses were conducted using TEM to further investigate the ionic and electronic transport mechanisms in the films, as depicted in Figure 6 and Supporting Information S1: Figures S8–S11. The images reveal a biphase structure

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TABLE 1 | Transistor parameters of vOECTs based on polymers and polymer: DtFGDA blends.

Materials	I_{on} (A)	I_{off} (A)	$g_{m,peak}$ (mS)	τ_{on} (ms)	τ_{off} (ms)	V_{on} (V)	Hysteresis (V)
P3HT	$(2.5 \pm 0.6) \times 10^{-2}$	$(2.0 \pm 1.3) \times 10^{-6}$	69.5 ± 21.0	266.7 ± 15.5	250.6 ± 9.1	−0.40 ± 0.02	0.42 ± 0.05
P3HT: DtFGDA	$(2.7 \pm 0.9) \times 10^{-2}$	$(4.7 \pm 1.8) \times 10^{-8}$	131.0 ± 46.9	79.1 ± 4.68	5.42 ± 0.99	−0.23 ± 0.03	0.04 ± 0.01
PQT	$(5.4 \pm 3.2) \times 10^{-4}$	$(1.7 \pm 0.7) \times 10^{-6}$	2.2 ± 1.1	1124.3 ± 51.9	245.9 ± 38.2	−0.72 ± 0.03	0.41 ± 0.04
PQT: DtFGDA	$(3.6 \pm 0.3) \times 10^{-2}$	$(2.6 \pm 1.8) \times 10^{-9}$	160 ± 10.4	215.8 ± 8.7	2.61 ± 0.44	−0.38 ± 0.02	0.13 ± 0.01
PBTTT	$(1.3 \pm 0.3) \times 10^{-3}$	$(2.4 \pm 0.5) \times 10^{-6}$	5.9 ± 1.6	531.5 ± 27.6	245.7 ± 70.1	−0.58 ± 0.03	0.26 ± 0.04
PBTTT: DtFGDA	$(6.7 \pm 0.1) \times 10^{-2}$	$(1.3 \pm 0.5) \times 10^{-8}$	230.9 ± 10.3	94.8 ± 2.1	16.7 ± 2.1	−0.21 ± 0.01	0.11 ± 0.02
DPPDTT	$(4.7 \pm 2.5) \times 10^{-6}$	$(4.7 \pm 1.4) \times 10^{-6}$	$(1.2 \pm 0.5) \times 10^{-5}$	7206.4 ± 187.4	1232.6 ± 138.3	−0.35 ± 0.03	0.25 ± 0.08
DPPDTT: DtFGDA	$(1.6 \pm 0.1) \times 10^{-3}$	$(1.2 \pm 0.5) \times 10^{-8}$	10.7 ± 6.7	363.3 ± 17.7	112.5 ± 7.5	−0.48 ± 0.04	0.12 ± 0.06

characterized by interspersed light and dark domains. Energy dispersive spectroscopy (EDS) was used to determine the composition of each domain (Figure 6A). The presence of carbon (C) throughout the film confirms its continuity, indicating that the dark domains are not voids (Supporting Information S1: Figures S8–S11). Additionally, sulfur (S) and fluorine (F) are identified elements unique to the polymer and crosslinker, respectively. The results show that the S distribution aligns well with the light domains in the morphology image, confirming that these are hydrophobic polymers. On the other hand, fluorine predominantly marks the dark domains (Figure 6B). It indicates that the excess crosslinker forms distinct domains through self-crosslinking, constructing a hydrophilic region in the film (Figure 6C). Combining the transfer curve and spectroelectrochemical measurements, different from neat films (Figure 6D), during gate bias stress, ions preferentially transport through the hydrophilic DtFGDA domains rather than the hydrophobic polymer domains (Figure 6E). Consequently, the hydrophobic polymer cannot be fully polarized. The interface forms between DtFGDA domains and hydrophobic polymer domains, resulting in field-effect doping at various interfaces in the blended films. Although ions do not directly enter the polymer domains, the extensive interface significantly enhances performance metrics such as transconductance, transient time, and hysteresis. Based on the TEM characterization, the mechanism is inferred through which the ethylene glycol photocrosslinker facilitates ionic-electronic synergistic conduction in the blended hydrophobic polymers. In this system, ions and electrons transport in different domains, which we define as ionic-electronic separation conduction that has never been reported in OECTs before. The existence of rough morphology and phase separation in the crosslinked films can be attributed to the Marangoni-like effect that arises during the solvent evaporation [53]. Initially, a homogeneous solution of hydrophobic polymer and hydrophilic crosslinker in chlorobenzene is spin-coated onto the substrate, resulting in the forming a single-phase transient wetting layer. During the solvent evaporation process, the polymer will solidify preferentially due to its lower solubility compared to the crosslinker. As more solvent evaporates, a polymer-rich phase and a crosslinker-rich phase are formed. The concentration gradient and the surface tension difference between the two phases, established throughout the transient wetting layer, induce instability at the polymer-crosslinker interface [54]. This initiates a Marangoni-like flow, ultimately causing phase separation and the formation of a rough morphology. The significant phase separation observed in four blended hydrophobic polymers demonstrates the broad applicability of this strategy for constructing ionic pathways across various hydrophobic systems. Despite significant improvements in transistor performance, there is a need to develop more robust crosslinked systems to further enhance the stability of separation conduction during operation (Supporting Information: Figure S12) and facilitate the translation of this technology into practical applications. Overall, using a hydrophilic crosslinker to enhance OECT performance is a promising, simple, and effective approach.

4 | Conclusion

In summary, a hydrophilic double-end trifluoromethyl glycol diazirine, DtFGDA, is newly developed, which is applied in vOECTs after being blended with a series of classical p-type

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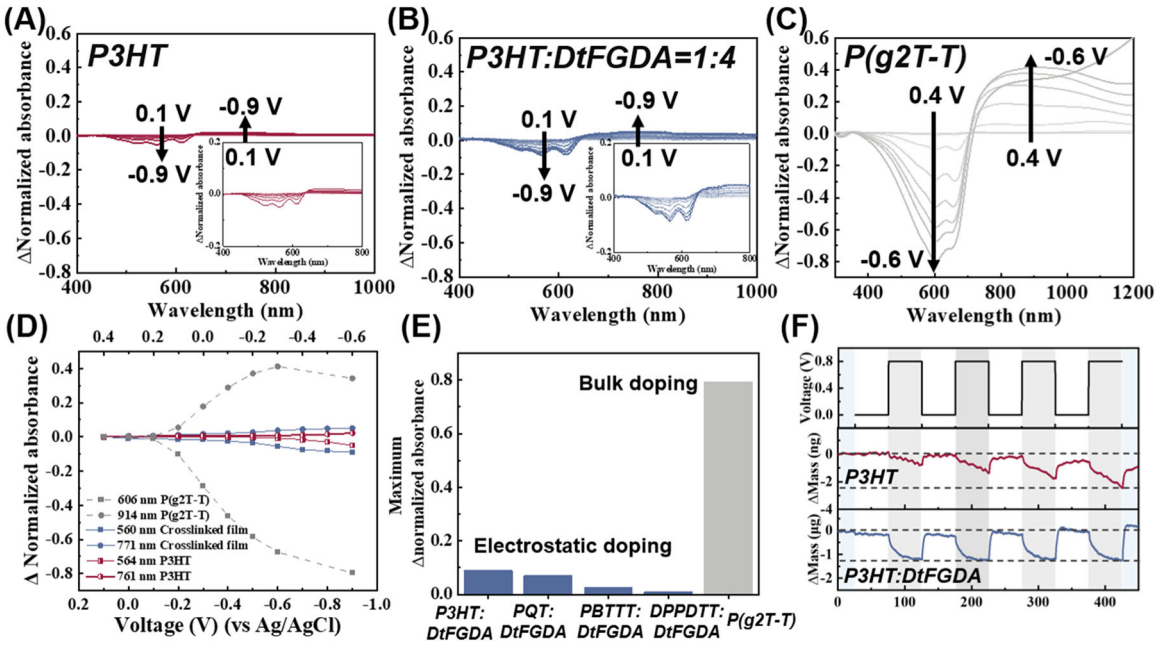


FIGURE 5 | Changes of normalized UV-Vis absorption spectra of P3HT (A), P3HT: DtFGDA (B) and P(g2T-T) (C) (Reproduced with permission: Copyright 2024, RSC. [51]). Dependencies of π-π* peak and polaron peak in UV-vis absorption spectra on the different film (D). Dependencies of π-π* peak in UV-vis absorption spectra on the different polymer with DtFGDA (E). Electrochemical quartz crystal microbalance (E-QCM) measurement of P3HT and P3HT: DtFGDA at 0–0.8 V with 0.1 mol/L PBS (F).

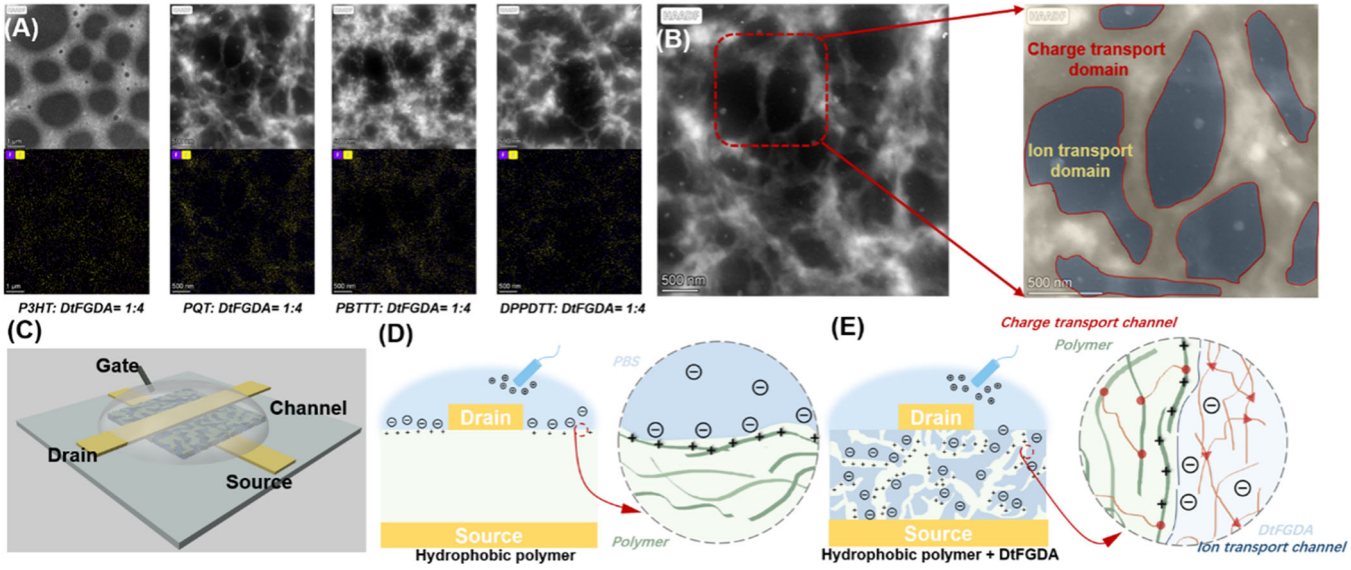


FIGURE 6 | (A) TEM morphological images (up) and EDS (down) of the crosslinked polymer: DtFGDA (1:4) films (B) Enlarged TEM morphological of the crosslinked PQT: DtFGDA (1:4) films. (C) Transistor channel composed of DtFGDA and hydrophobic polymers. Schematic working mechanism of neat film (D) and crosslinked film (E) in transistor. To facilitate clearer observation, the contrast of the EDS was increased.

hydrophobic semiconducting polymers (P3HT, PQT, PBTTT, and DPPDTT). We introduce a hydrophilic chain into the photocrosslinker molecule, which is essential for direct photolithography, and blend it with hydrophobic polymers, utilizing the synergistic and complementary effects of the different components to meet the requirements of OMIECs. By this strategy, the ionic injection/ejection in the blended film is promoted, achieving high-resolution patterning and high device performance. Notably, based on the characterization of SEC and TEM, a unique conduction mechanism of OMIECs is revealed in this work. Specifically, within the crosslinked films, ions mainly insert into the hydrophilic crosslinker-rich phase, while the electron transport is in the hydrophobic polymer-rich phase. We define this mechanism as ionic-electronic separated conduction. The distribution of hydrophilic crosslinker-rich phase throughout the blended films facilitates aqueous electrolyte insertion, enabling electrostatic doping of the hydrophobic polymer-rich phase. Consequently, after direct photolithography, the patterned films demonstrate outstanding electrochemical responses, including significantly enhanced

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transconductance, reduced turn-on time, and diminished hysteresis. Dual pathways are constructed in the channel through significant phase separation of excessive hydrophilic photocrosslinker and hydrophobic semiconducting polymer, with each pathway serving its specific function. The methodology introduced here offers a unique approach to constructing OMIECs, permitting independent regulation over the conductivity and ionic transport properties of the film. This finding provides an excellent and convenient platform for the development of OMIECs and guides future studies on blended semiconductor films in ionic-electronic coupling devices.

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Conflicts of Interest

The authors declare no conflicts of interest.

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Supporting Information

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